

# Majorana Modes in the Machine

Week 5: Transitioning to Quantum Circuit Measurement

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Advanced Seminar — AI-Augmented Theoretical Physics

June 19, 2026

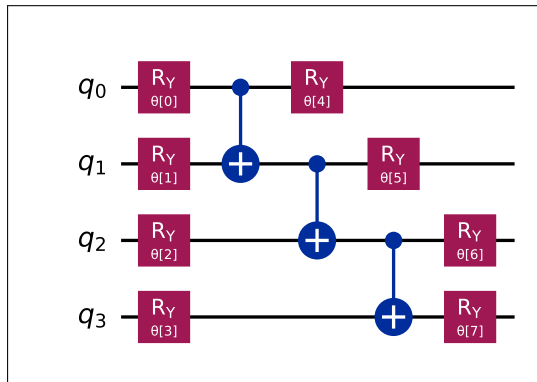
## Week 5 Objectives: Moving to Block 3

- **Main Goal:** Shift from "Exact Matrix Math" to "Gate-Based Simulation."
- **Task 1:** Ground State Preparation via VQE (Variational Quantum Eigensolver).
- **Task 2:** Defining the Non-local String Order Parameter  $\mathcal{O}_{string}$ .
- **Task 3:** Mapping Majorana Observables to Qubit Measurement Gates.
- **Task 4:** Numerical Evidence.

# Task 1: VQE Circuit Ansatz (Visual)

## Hardware-Efficient Ansatz:

- Uses only  $R_y$  to keep the state in the real domain.
- Linear CNOT entanglement mimics 1D hopping.



## $R_y$ Gate

For  $L = 4$ , the configuration space consists of  $2^4 = 16$  possible states:

$$|\Psi\rangle = c_0|0000\rangle + c_1|0001\rangle + \cdots + c_{15}|1111\rangle$$

Initially,  $|\psi\rangle = [1, 0, 0, \dots, 0]^T$ . An  $R_y(\theta)$  gate on  $q_0$  performs:

$$R_y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle$$

In the 16D vector, the value "1" splits:

- Index 0 ( $|0000\rangle$ ): becomes  $\cos(\theta/2)$
- Index 8 ( $|1000\rangle$ ): becomes  $\sin(\theta/2)$

# CNOT Gate

Entanglement occurs when a gate moves amplitudes between binary configurations.

## CNOT(0,1) Operation

If  $q_0$  is 1, flip  $q_1$ . In the vector:

- 1  $|1000\rangle$  (Index 8) satisfies the condition.
- 2 It is transformed into  $|1100\rangle$  (Index 12).

Physical meaning: isolated excitations are combined into **paired states**, building the topological correlations needed for Majoranas.

# Hilbert Space Evolution: Mathematical Verification

**Mathematical Constraint:** Post-entanglement states cannot be factored into independent single-qubit probabilities. The evolution must be tracked across the full  $2^4 = 16$  dimensional state vector.

**State Vector Evolution Table (Tracking non-zero amplitudes):**

| Index (Dec) | Basis ( $q_0q_1q_2q_3$ ) | Init State $ \psi_0\rangle$ | Post $R_y(\theta_0)$ $ \psi_1\rangle$ | Post CNOT <sub>0→1</sub> $ \psi_2\rangle$ |
|-------------|--------------------------|-----------------------------|---------------------------------------|-------------------------------------------|
| 0           | $ 0000\rangle$           | 1                           | $\cos(\frac{\theta_0}{2})$            | $\cos(\frac{\theta_0}{2})$                |
| 1           | $ 0001\rangle$           | 0                           | 0                                     | 0                                         |
| ⋮           | ⋮                        | ⋮                           | ⋮                                     | ⋮                                         |
| 8           | $ 1000\rangle$           | 0                           | $\sin(\frac{\theta_0}{2})$            | <b>0</b> ( <i>Amplitude shifted</i> )     |
| ⋮           | ⋮                        | ⋮                           | ⋮                                     | ⋮                                         |
| 12          | $ 1100\rangle$           | 0                           | 0                                     | $\sin(\frac{\theta_0}{2})$                |

# Hamiltonian Simplification: The Pure YY Limit

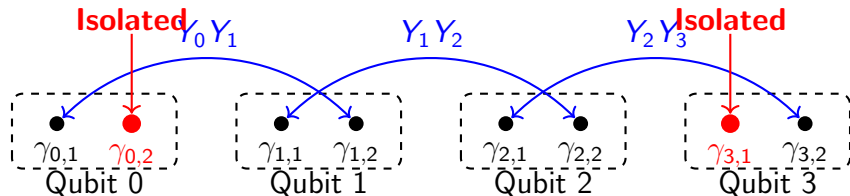
## General Qubit Mapping (Block 2):

$$H_K = -\frac{\mu}{2} \sum Z_j + \frac{t-\Delta}{2} \sum X_j X_{j+1} + \frac{t+\Delta}{2} \sum Y_j Y_{j+1}$$

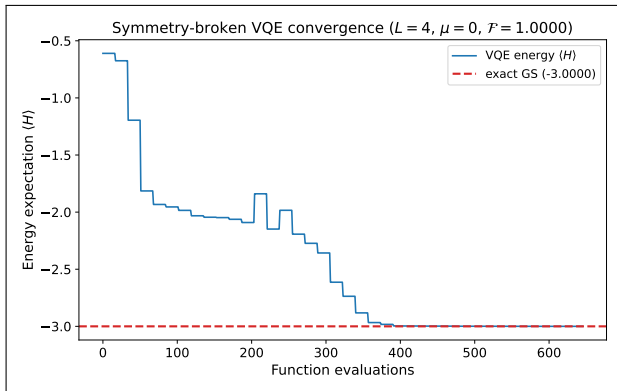
## Topological Sweet Spot ( $t = \Delta$ , $\mu = 0$ ):

$$\mathbf{H} = \sum_{j=0}^{L-2} \mathbf{Y}_j \mathbf{Y}_{j+1} \equiv i \sum_{j=0}^{L-2} \gamma_{j,1} \gamma_{j+1,2}$$

- **Algebraic Cancellation:** Setting  $t = \Delta$  strictly eliminates all  $X_j X_{j+1}$  components.
- **Pairing Shift Geometry:** The pure  $Y_j Y_{j+1}$  interaction fundamentally alters the topology by coupling the **left mode** ( $\gamma_{j,1}$ ) to the **right mode** ( $\gamma_{j+1,2}$ ), effectively isolating the inner edge modes.



# Numerical Verification: VQE Convergence Dynamics



## Optimization Parameters:

- **Algorithm:** L-BFGS-B (Gradient-based).
- **Cost Function:**  
 $C(\vec{\theta}) = \langle H \rangle - \lambda \langle P \rangle$ .
- **Penalty ( $\lambda = 0.1$ ):** Enforces strictly even parity ( $P = +1$ ) selection, breaking topological degeneracy.

## Task 2: String Operator

### Boundary Majorana Correlator to Qubit Mapping:

$$\hat{O}_{corr} \propto i\gamma_{0,2}\gamma_{3,1} = i(Y_0)(Z_0Z_1Z_2X_3) = -X_0Z_1Z_2X_3$$

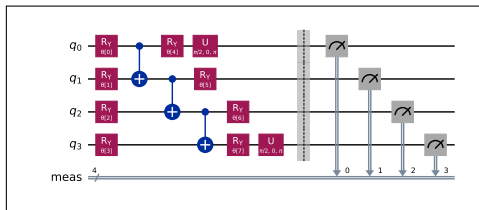
- **1. Breakdown of the Landau Paradigm:** The topological phase hosts no local symmetry breaking. All local single-qubit expectations strictly vanish:

$$\forall j \in \{0, 1, 2, 3\}, \quad \langle X_j \rangle = 0, \quad \langle Y_j \rangle = 0, \quad \langle Z_j \rangle = 0$$

Non-local order parameters are mandatory to detect this hidden order.

- **2. Ground State Commutation Constraint:** A saturated non-zero invariant requires  $[H, \hat{O}_{string}] = 0$ . The naive boundary operator  $\hat{O}_{naive} = X_0X_3$  anti-commutes with  $H$ , creating high-energy quasiparticle excitations and yielding  $\langle X_0X_3 \rangle \equiv 0$ .

# Task 3: Basis Rotation via Hadamard Transform



## Hardware Constraint:

Superconducting qubits are strictly restricted to  $Z$ -basis projection.  
Majoranas require  $X$ -basis detection.

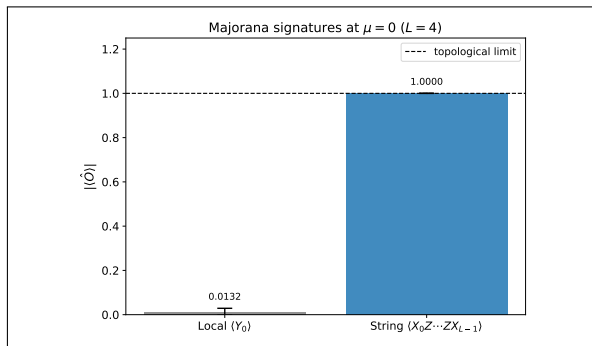
**Target Observable:**  $\hat{O}_{corr} = -X_0 Z_1 Z_2 X_3$

**Algebraic Mapping ( $HXH = Z$ ):**

$$\begin{aligned} HXH &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \equiv Z \end{aligned}$$

- $H$  is applied specifically to  $q_0$  and  $q_3$  to reconstruct the non-local string.

## Task 4: Numerical Evidence: The 1.0 Correlation



- **Left ( $Y_0$ ):**  $\approx 0$  due to anti-commutation with the global parity operator ( $\{Y_0, \hat{P}\} = 0$ ).
- **String ( $X_0 Z_1 Z_2 X_3$ ):** Exactly 1.0000 (Theoretical limit).
- **Algebraic Symmetry:** The commutation  $[H, \hat{O}_{corr}] = 0$  identifies this as a conserved topological invariant.
- **Result:** Non-local Topology confirmed in the gate-based model.

# Summary & Next Steps

- **Completed:** VQE ground state preparation.
- **Discovery:** The non-local string order parameter is a robust topological signature.
- **Next:** Sweep  $\mu$  to generate the full Phase Diagram using only quantum circuit results.